



Ramakrishna Mission Vivekananda Educational and Research Institute

PO Belur Math, Howrah, West Bengal 711 202

School of Mathematical Sciences

Department of Computer Science

BDA, CS and Math, Fall 2024, Mid-term examination

CS230: Machine Learning

Instructor: Br. Tamal Maharaj

Student Name (in block letters): **SOHAM BHATTACHARYA**

Student Reg. No: **B2430059**

Signature: *Soham Bhattacharya*

Date: 21 Sep 2024

Max Marks: 100

Time: 2 hrs

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Question 1: [Please keep your answers concise and to the point] (20 points)

1. (2 points) What are the three major categories of Machine Learning?
2. (2 points) What is out-of-core learning, and when is it typically used?
3. (2 points) What is the difference between a model parameter and a model hyperparameter? Provide an example for each.
4. (2 points) Which linear regression training **algorithms** are suitable when you have a training set with millions of features and which are not? No justification required.
5. (2 points) Can gradient descent get stuck in a local minimum when training a logistic regression model? Briefly justify your answer.
6. (2 points) Explain the No Free Lunch Theorem in the context of Machine Learning.
7. (2 points) What is stratified sampling, and what is its purpose?
8. (3 points) Assume that the input DataFrame you are working on is called **Salary** and has 'Bonus as one of its attributes'. Some entries for this attribute are missing. Write three different strategies to clean the data, using Python code.
9. (3 points) When fine-tuning a model, which search strategy is usually preferred: Grid Search or Randomized Search? Explain two benefits of the preferred choice.

Question 2 (20 points)

1. (5 points) Why is feature scaling important in machine learning? Provide two popular methods for performing feature scaling along with corresponding equations for each method.
2. (15 points) Express the linear regression model in matrix form, given n features and m data points, where the input matrix is X , the model parameters are Θ , and the output vector is Y . Specify their dimensions. Write the cost function and its gradient in matrix form. Finally, derive the normal equation for estimating Θ .

Question 3 (20 points)

1. (12 points) Write the normal equation for Ridge Regression. How does Ridge Regression address the invertibility problem in Linear Regression? Explain why Lasso Regression does not have a normal equation.
2. (8 points) Under what circumstances would you prefer to use:
 - (a) Lasso Regression over Ridge Regression?
 - (b) Elastic Net over Lasso Regression?

Question 4 (20 points)

1. (5 points) Plot the Sigmoid function, $\sigma(x)$, and compute its derivative with respect to x in terms of $\sigma(x)$.
2. (10 points) What are the various categories of multiclass classification models? For K classes, how many classifiers are trained in each category?
3. (5 points) Write the Cross-Entropy cost function and demonstrate how it generalizes the Logistic Regression cost function.

Question 5 (20 points)

1. (2*7 + 6*1 = 20 points) Write down the output of the following code snippets directly on this question paper. Assume that numpy and pandas are already imported as np and pd, respectively.

```
b = np.arange(24).reshape(6, 4)
print(b[1:2, :])
```

[4, 5, 6, 7]

```
x = np.array([[1, 2, 9], [4, 5, 6]])
print(np.max(x, axis=1))
```

[4, 5, 9]

```
x = np.arange(1, 10).reshape(3, 3)
print(np.sum(x, axis=0))
```

[6, 15, 24]

```
g = np.arange(24)
g2 = g.reshape(4, 6)
print(g.ndim)
print(g2.ndim)
```

(1)

(4, 6)

```
m = np.array([20, -5, 30, 40])
print(m < [15, 16, 35, 36])
```

False

```
m = np.array([20, -5, 30, 40])
print(m[m < 25])
```

[True, True, False, False]

```
a = np.array([1, 5, 3, 19, 13, 7, 3])
print(a[1:2])
print(a[1::2])
```

[5]

[5, 19, 7]

```
people = pd.DataFrame({
    "birthyear": {"alice": 1985, "bob": 1984, "charles": 1992},
    "hobby": {"alice": "Biking", "bob": "Dancing"},
    "weight": {"alice": 68, "bob": 83, "charles": 112},
    "children": {"bob": 3, "charles": 0}
})
print(people.iloc[2]) # 3 pts
```

{ "alice": "Biking", "bob": "Dancing" }

```
print(people[people["weight"] > 80]) # 3 pts
```